

Fusion of Multi-Sensor-Based Biomechanical Gait Analysis Using Vision and Wearable Sensor

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Abstract—The purpose of this research work is to investigate the biomechanics of pelvis, hip, knee, and ankle joint motion using a Kinect sensor & inertial measurement unit (IMU) sensor during the normal walk. In this paper, a very cost-effective gait analysis system based on Microsoft Kinect v2 and Inertial Measurement Unit (IMU) device is presented. Kinect sensor is used for acquiring 3D skeleton data (camera (x, y, z), depth (x, y) orientation (x, y, z, w), color (x, y)) with 25 human body joints. For this analysis, the lower extremities joints i.e. spinal cord joint, hip, knee, and ankle joints of both left and right legs are being considered. The main contribution of this research work is the joint angle calculation of lower extremities of human gait based on Microsoft Kinect sensor V2 and IMU sensor. From the law of cosine, the joint angle is calculated between the two joints and plotted for a single subject. We came with the observation that the characteristics of the human knee joint and ankle joint are inversely related to each other. There are two sharp humps for knee and ankle joints during the normal walk. During the swing phase, the knee joint is highly activated while during toe-off and heel strike it is least activated. This analysis of clinical data is very useful for prosthesis limb and exoskeleton design. The stability of calculated joints trajectories is validated using the limit cycle curve. A system is designed for real-time analysis of biomechanics of different lower limbs joints using gait.

Index Terms—Biomechanics, gait analysis, humanoid robot, sensor, inverse kinematics.



I. INTRODUCTION

A. Overview

HUMAN gait motion analysis is a field of great interest. The human gait is the manifestation of change in the joint angles of the hip, knee, and ankle. It is a complex learning process which involves various body part including the human brain. The human brain is considered as a central nervous

system that is responsible for reflexive behavior [1]. The left and right lobes of the human brain play a crucial role in the control of activities of human motion of the left and right leg. The left hemisphere is more active for a right-handed person and the right hemisphere is more active for left-handed person activities [2]. This analysis presents the important conclusion in understating the learning mechanism of the brain and problem faces by elderly and differently-abled person [3]. The brain has an enormous capability regarding control of our day-to-day activities [4]. This is the motivation to understand the bio-mechanism of the brain for human walking [5]. The walk is a reactive behavior that is tightly coupled between perception and action. The Cerebellum is the main part of the brain which is responsible for body control and correct the voluntary movement of the body. It is also responsible for learning various body motions [6]. The neurological disorder and sensorimotor deficits are the major reason for larger gait variability in patients suffering from Parkinson and Stroke [7]. The Human gait has great variability due to many factors namely environment, sensorimotor noise, and time-varying mechanics of muscles (due to e.g. fatigue, over stress) that influence our walking [8]. It is responsible for the change in body posture and motor action [9]. It also explains the bio-mechanism of the brain & motor interaction [10].

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B. Signification of Work

The Traditional approaches are unable to deal with the high variability between a repeated or periodic gait movement [11]. Here, In this research, we have explored a data-driven modeling technique to estimate joint angle value through a Kinect sensor. The proposed research work is designed to estimate the biomechanics of knee and ankle joints during all phases of gait. The Joint stiffness increases with muscle activation. Due to this muscle movement varies during different walk [12]. One of the main fields is biomedical which is most prospective and promising for upcoming times [13]. As the age grows people face the common problems of upper and lower extremities due to aging of the neuron, which impacts the health physically and mentally [12]. There are generally common diseases like stroke, cerebral palsy [14], Parkinson, etc. which impact on motor functions of aging persons. Thus, it is necessary to monitor, track and analyze the locomotion of these elderly peoples with the help of wearable sensor [15] or low-cost advanced vision-based sensors without disturbing or interrupting their free-living activities [16], [17]. Real-time human motion tracking can be applied to several biomedical applications, such as clinical gait analysis, rehabilitation and joint motion analysis etc [18].

C. Author's Contributions

There are three major contributions of this work is listed here:

- 1) **Database collection:** Total 22 subject's skeleton data is collected through kinect sensor and IMU sensor. The subject was considered of the of age group 18 to 45 years including 12 males and 10 females.
- 2) **Feature Extraction:** Analyses of the features is done i.e. joint positions (F_j), angular vector (F_a), velocity vector (F_v) and distance (F_d).
- 3) **Joint Angle Calculation:** We calculated the Joint angles of the hip, knee, and ankle using the proposed algorithm 11. We plotted the joint angle trajectories of knee and ankle and deduce the hypothesis which was proposed. The motion analysis was done by analyzing relative joint positions, velocity, and angular velocity of the skeleton In the second phase, we collected the gait data of 12 subjects using inertial measurement uni(IMU) and solve the inverse kinematic using the proposed algorithm 2 for calculation of the values of the joint angles.
- 4) **Trajectories plotting and stability analysis:** Finally, the joint angle trajectories of hip, knee, and ankle is plotted and validated using phase plots of both modality. Fig. 8 shows the phase plots for different joints.

D. Organization of the Paper

The remainder section of this paper is structured as follows. The second section is about the literature review about gait data analysis using kinect & IMU sensors. Section 3 the describes of data collection and preprocessing of raw data. This section also describe the methodology used in proposed work. Section 4 is about results and analysis. This section also

Algorithm 1 Proposed Joint Angles Calculation Algorithm $\theta(n)$, Where, θ_{LH} Is for Left Hip, $\theta_{RH}[i]$ Is for Right Hip, $\theta_{LK}[i]$ Is for Left Knee, $\theta_{RK}[i]$ Is for Right Knee, $\theta_{LA}[i]$ Is for Left Ankle, and $\theta_{RA}[i]$ Is for Right Ankle

Result: Joint Angle values:
1 Input: $Position(joint_i)$, $Position(joint_j)$ & $Position(joint_k)$
2 Output: $\{\theta_{LH}[i], \theta_{RH}[i], \theta_{LK}[i], \theta_{RK}[i], \theta_{LA}[i], \& \theta_{RA}[i]\}$
3 Procedure
4 Initialization: $d(i, j) \leftarrow 0, d(j, k) \leftarrow 0, d(i, k) \leftarrow 0$.
5 Calculate euclidean distance $d(i, j)$:

$$d(i, j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2} \quad (1)$$

6 Calculate euclidean distance $d(j, k)$:

$$d(j, k) = \sqrt{(x_j - x_k)^2 + (y_j - y_k)^2 + (z_j - z_k)^2} \quad (2)$$

7 Calculate euclidean distance $d(i, k)$:

$$d(i, k) = \sqrt{(x_i - x_k)^2 + (y_i - y_k)^2 + (z_i - z_k)^2} \quad (3)$$

8 while $((i, j, k) < n)$ **do**
9 Calculate the cosine joint angle values:

$$\cos \theta = \frac{(d(i, j)^2 + (d(j, k))^2 - (d(i, k))^2)}{(2 * d(i, j) * d(j, k))} \quad (4)$$

10 Calculate the joint angle θ :

$$\theta = \cos^{-1} \frac{(d(i, j)^2 + (d(j, k))^2 - (d(i, k))^2)}{(2 * d(i, j) * d(j, k))} \quad (5)$$

11 end



Fig. 1. Data acquisition using kinect in our laboratory.

present the stability analysis. The final section 5 describes our conclusion and future scope.

II. LITERATURE REVIEW OF RELATED WORKS

Gabel. et al [19], [20] proposed to use Kinect for human body analysis as a markerless and less expensive, and non-intrusive method. Their proposed technique is to be used for the continuous tracking of gait at home. Obdrzalek *et al.* [21] worked in the field of pose estimation and achieved great accuracy as compared to commercially available motion capture system also deduce that Kinect feels occlusion and it severely affects accuracy. In this experiment, he used a first-generation Kinect. The second-generation Kinect was introduced in 2014 and it uses the time of

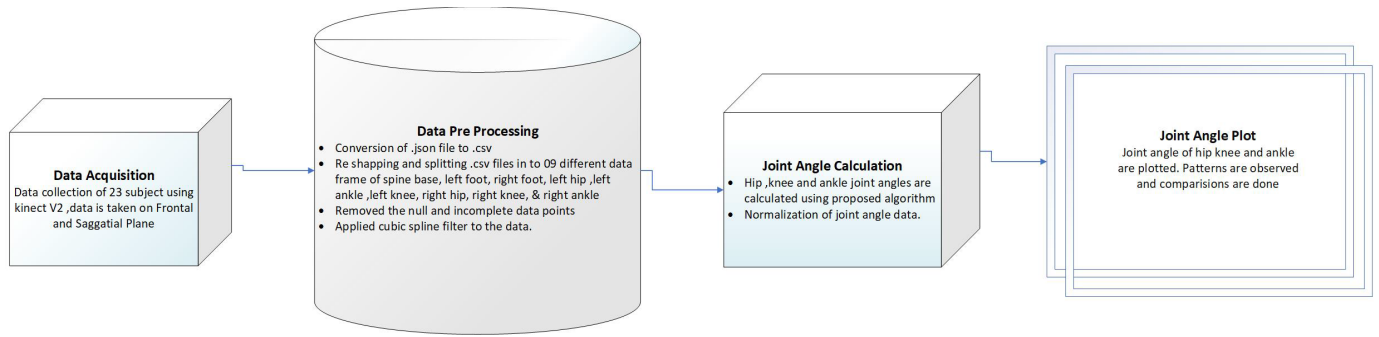


Fig. 2. Process Flow Diagram.

Algorithm 2 Inverse Kinematics Solution to Generate the Joint Angle Values:

Result: Joint Angle values: $\{\theta_{LH}[i], \theta_{RH}[i], \theta_{LK}[i], \theta_{RK}[i], \theta_{LA}[i], \theta_{RA}[i]\}$

1 **Initialization 1:** $t_c \leftarrow 0.0167$, $l_s \leftarrow 4$, p_i , where $i \leftarrow 1, 2, 3, \dots, n$;

2 **Initialization 2:** $Per \leftarrow \{10, 30, 50, 60, 73, 87, 100\} / 100$;

3 **Initialization 3:** $t_{lrr} \leftarrow t * per [1]$, $t_{mst} \leftarrow t * per [2]$, $t_{ts} \leftarrow t * per [3]$, $t_{ps} \leftarrow t * per [4]$, $t_{is} \leftarrow t * per [5]$, $t_{msw} \leftarrow t * per [6]$, $t_{tsw} \leftarrow t * per [7]$;

4 **Initialization 4:** $t_{sp} \leftarrow \{t_0, l_r, m_{st}, t_s, p_s, i_s, m_{sw}, t_{sw}\}$, $j \leftarrow 0$, $k \leftarrow 0$, $\theta_K = [\theta_{LH}, \theta_{RH}, \theta_{LK}, \theta_{RK}, \theta_{LA}, \theta_{RA}]$;

5 **while** While condition **do**

6 $tmp[i] = p_i * x^i + er$, where, $i=1,2,\dots,n$, n -degree of polynomial ;

7 $j=j+1$;

8 **end**

flight technology for accurate estimation of depth image accuracy also it increased the number of body joints to 25. Latorre *et al.* [22] used Kinect v2 and calculate spatiotemporal gait parameters such as step length, velocity, stride time, and kinematics parameters like flexions of the knee, ankle, and hip.

As the Kinect is extensively being used in gait analysis of elderly persons suffered from various gait disorders thus various research has been done in this direction. Lange *et al.* [23] used a game-based rehabilitation tool (Using kinectV1) to improve the balance in adults after neurological injuries [24]. Saini *et al.* [25] presented a novel framework of continuous human activity recognition using Kinect and classified the activities using deep learning methods. Semwal *et al.* [26], [27] on the other side presented a novel method of joint angle calculation based on the IMU sensor's accelerometer data using inverse kinematics. Chakraborty and Nandy [28] proposed a gait phase detection using multiple Kinect and achieved outstanding 96.92% accuracy. Ince *et al.* [29] proposed a novel biometric system that can detect human activities in 3D space. They obtained joint angle features using an RGB depth camera and used KNN for activity recognition. Nandy *et al.* 2015 [30] implemented IGOD suite for capturing data for gait data used a unique biometric signature for recognition also used Naïve Bayes classifiers for classification.

III. PROPOSED METHODOLOGY & TRAJECTORIES GENERATION

We used a low-cost Microsoft Kinect V2-based human motion capturing system for real-time human motion capture.

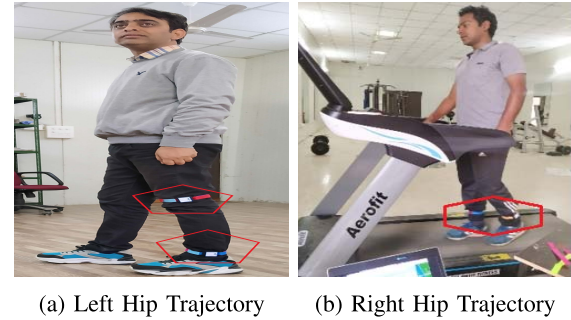


Fig. 3. IMU sensor placed on different subject.

Kinect V2 can track 6 skeletons at a time and each skeleton represented by Kinect has 25 joints. Joints are represented by numbers (0-24). Every joint has 11 different properties: color (x, y); depth (x, y); Camera (x, y, z); and orientation (x, y, z, w). Kinect has two cameras (i) Colour RGB of resolution 1920×1080 and (ii) Depth camera of 512×424 . The depth range of Kinect is about eight meters but the skeleton capture range is 0.5meter to 4.5 meters. The recording rate of data through Kinect was 30fps. Kinect also used quaternions for representing joint orientations. Quaternions are 4D ways to store the 3D orientations. On the other hand, an Inertial measurement unit sensor consisting of an accelerometer, gyroscope, and magnetometer is very popular in cost-effective GAIT analysis and for finding out pathological gait. IMU BWT61CL-E is being used for our analysis purpose. Accelerometer data is used to identify joint positions and ultimately the joint angle measurement of lower extremities.

Fig. 7 shows the observed joint angle trajectories of hip and knee using IMU sensor in multi-view. Fig 8 shows the limit cycle curve. The limit cycles shows that generated trajectories of Knee and ankle joints are stable,

A. Data Acquisition Using Kinect Sensor

Total 22 subjects of different age groups(18-45Years) Male and Female are taken for the gait analysis. One Kinect sensor is placed on the frontal plane and another on the sagittal plane for the data recording. The Kinect is placed at a height of 1.5 feet above the floor so that it may properly detect the skeleton of the subject and for clear data recordings. The Subject is trained to walk on the treadmill at a very normal speed of 1 mile/h, we may assume that it is a normal slow



Fig. 4. Observed Joint angle trajectories of hip using kinect in multi-view.

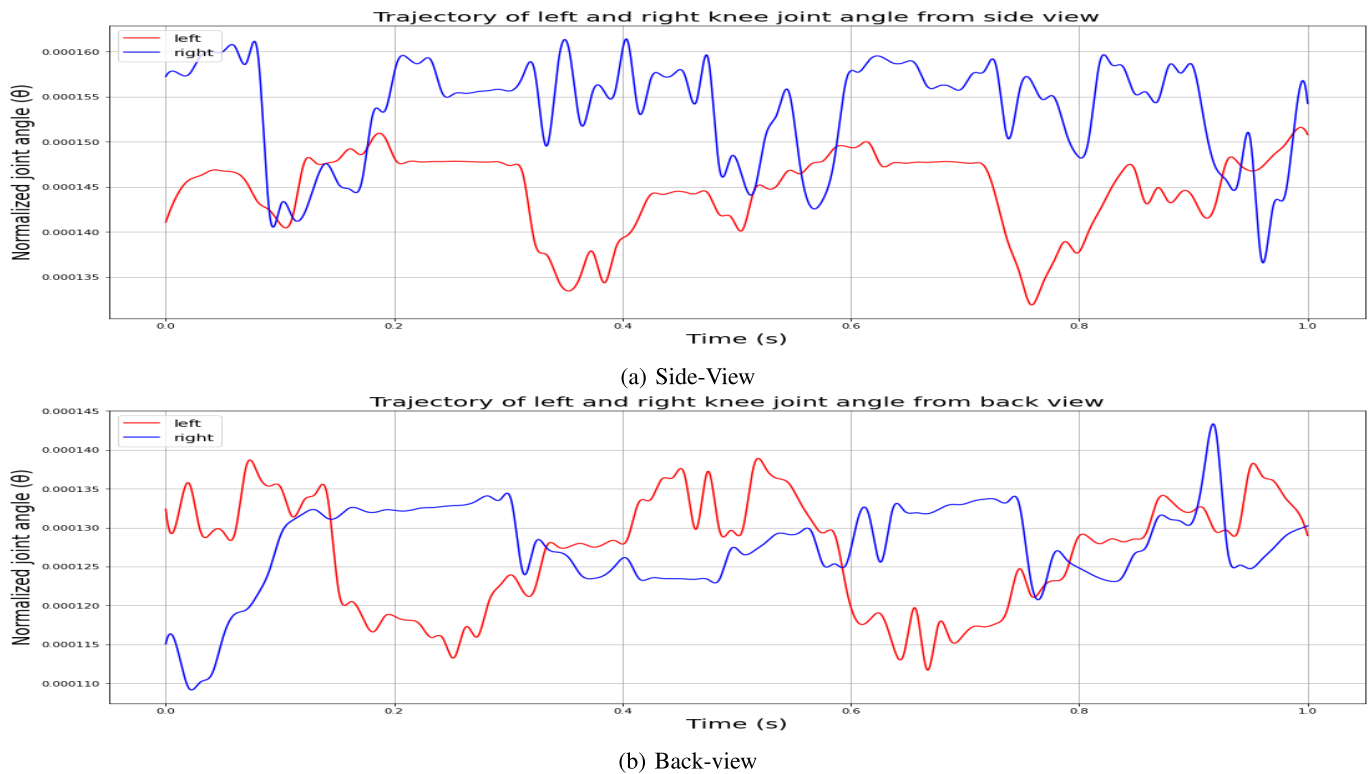


Fig. 5. Observed Joint angle trajectories of knee using kinect in multi-view.

walk. The Kinect gives us the output of 25 Skeleton joints of the human body. For our analysis of gait, we deal with the joint of spine base, hip, knee and ankle left and right leg. The whole recording of the normal walk is taken in a

controlled manner and a controlled environment of a Human motion capture laboratory. As shown in Fig. 1.

The output data we got after recording are camera (x,y,z), depth (x,y) orientation (x,y,z,w), colour (x,y). The data is

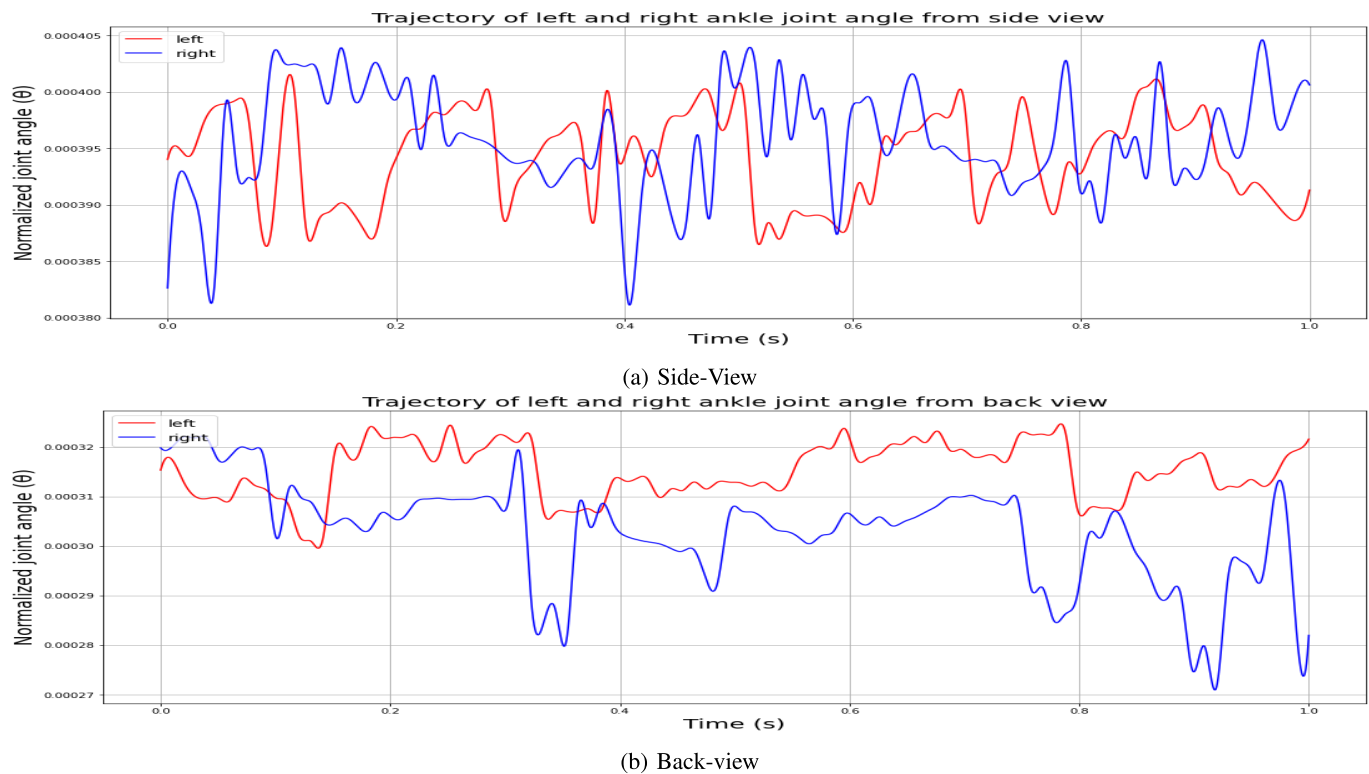


Fig. 6. Observed Joint angle trajectories of ankle Joint using Kinect in multi-view.

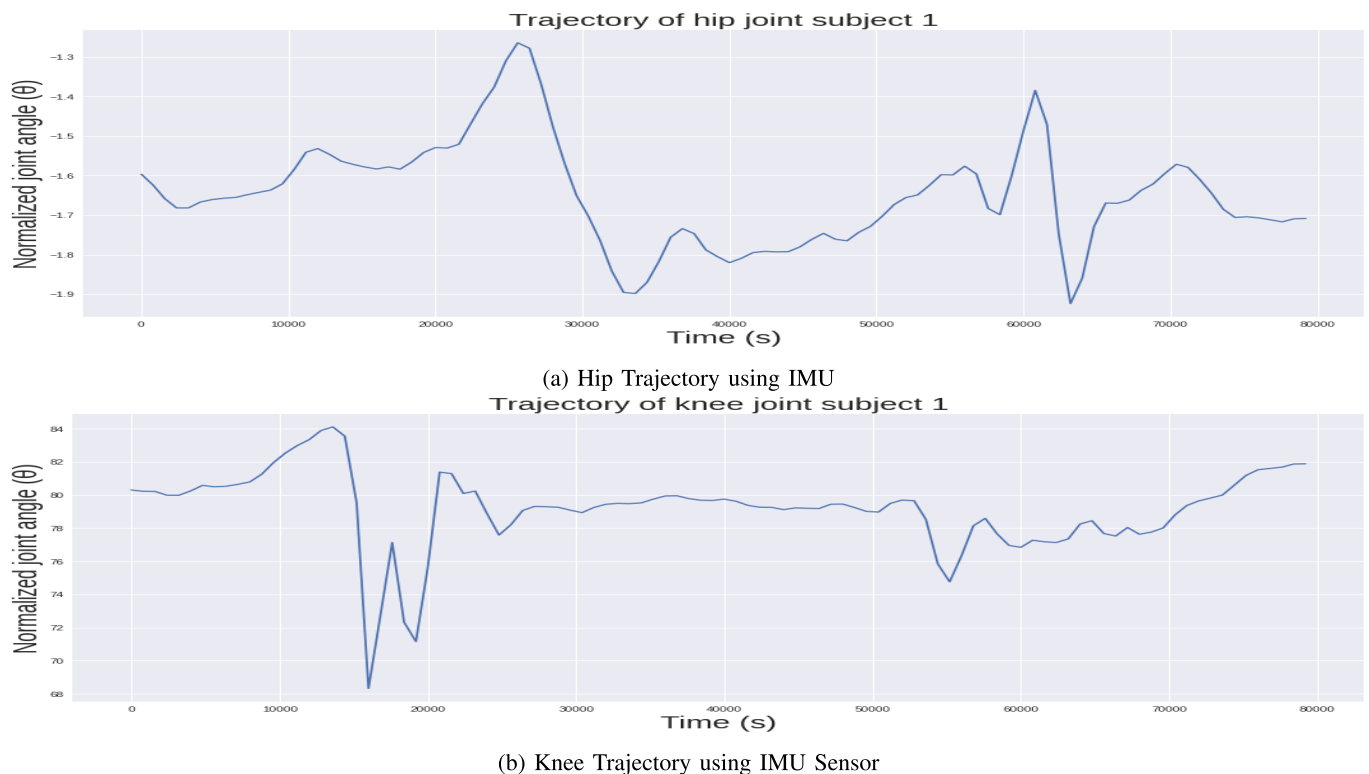


Fig. 7. Observed Joint angle trajectories of hip and knee using IMU sensor in multi-view.

collected in.json files. We pre-process the data and remove the noise from the extracted data. For our analysis purpose, we take the camera (x, y, z) into account and calculated the 2 dof joint angles of hip, knee, and ankle based upon the algorithm.

B. Calculation of Joint Angles and Their Trajectories Using Kinect

The algorithm 3 is used for joint angle calculation of hip, knee, and ankle. It is important to note that for calculation of the joint angle we have to consider the nearest coordinates of

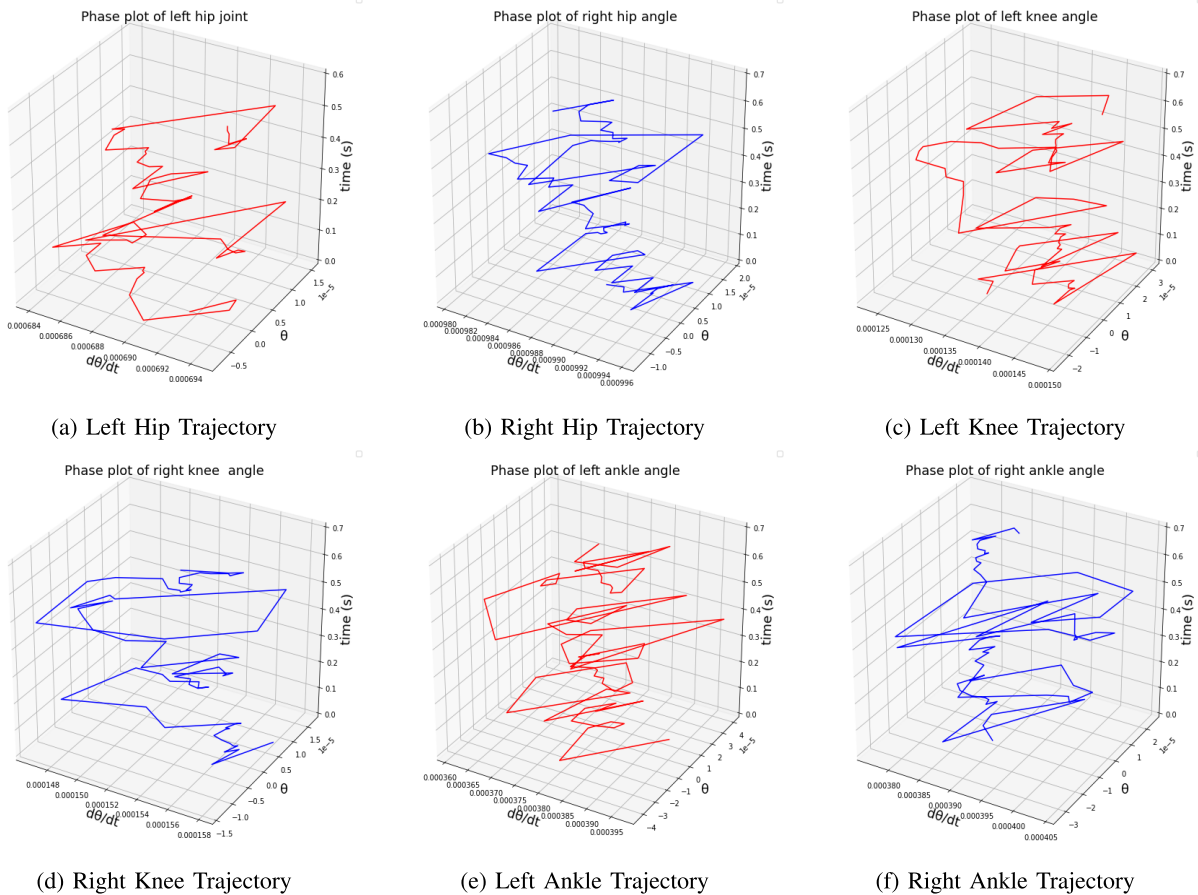


Fig. 8. Phase plots of different joints.

Algorithm 3 Calculation of Joints Trajectories for Shank, Thigh, Hip, Knee, Ankle and Foot Using Inverse Kinematic

Result: Joint angle trajectories for Shank, Thigh, Hip, Knee, Ankle & Foot

- 1 **Input:** Raw data collected through accelerometer X, Y, L1, L2;
- 2 **Output:** Joint Angles;
- 3 **Procedure;**
- 4 Solve the inverse kinematic for co-ordinate(x,y);
- 5 Calculate the Joint Angle values ;
- 6 Plot the curve;

joints. For example, in the calculation of the joint angle of the knee, we should have the known coordinates of adjacent joints i.e hip and ankle. The Complete flow diagram of our process is shown below in Fig. 2

C. Spatio Temporal Parameters Measurement Using IMU Sensor

In the first step of data acquisition using an IMU sensor, 12 subjects (07 male and 05 female) of different age range (18-45 years) are used for data recording. IMU sensor is placed at 03 different positions of hip, knee, and ankle of the left and right leg. IMU sensor [31], [32] consist of 03 sensors i.e. accelerometer, gyroscope, magnetometer so we got the three output (a_x, a_y, a_z), ($\omega_x, \omega_y, \omega_z$) and (m_x, m_y, m_z). The accelerometer data is used in our analysis.

As our IMU sensor (BWT 61CL-E) consists of an inbuilt Kalman filter, so the raw data is filtered by the Kalman filter. In the second step, the data is pre-processed, removed the nan values, and cleaning of the raw data. In this step we, convert.txt file to.csv for our analysis purpose.

D. Joint Angle Calculation Using IMU

The method proposed is based on an affordable sensing system using Inertial measurement unit [33], [34]. For evaluating the joint angle of the Right leg Knee, two IMU(BWT6CL-E) containing accelerometer and gyroscope were attached to Shank (just above the ankle) and at the end of the Thigh (near hip region) of the right leg. Velcro strips were used for attaching the IMU on the given positions on tight trousers for providing a firm attachment to the segments. As IMU is connected to the laptop through Bluetooth signaling, thus continuous 3-D readings were recorded of acceleration and gyroscope on the system using software BWT mini IMU at a frequency of 100 Hz. Thus, we have acceleration values of thigh and shank. Positions (x, y, z) were calculated by double integration's of acceleration values (a_x, a_y, a_z). Now the proposed algorithm 11 and algorithm 2 are given for calculating the joint angles.

IV. RESULTS & DISCUSSIONS

The two approaches were used to analyze the gait hip, knee, and ankle pattern. In a computer-vision-based study,

we analyze the gait on the frontal and sagittal plane i.e multi-view and it is also seen the behavior of hip, knee, and ankle using the Inertial measurement unit place on the subject's body. Finally, the stability measurement is done by plotting the trajectories of θ versus $\dot{\theta}$ which is a spiral type curve, shows that the walk was stable, where θ is joint angle and $\dot{\theta}$ is the differentiation of joint angle theta. It was observed that during the normal walk, in ideal form the trajectories of hip joints are sinusoidal due to oscillatory motion in nature. The knee and ankle joints have two inversely related sharp humps. During toe-off and heel strike, the knee joint has the least actuation, whereas ankle joints have maximum actuation during the double support phase or stance phase. Fig. 4, Fig. 5, Fig. 6 shows the observed Joint angle trajectories of hip, knee, and ankle joints respectively using Kinect in multi-view. It shows the knee and ankle joints are inversely related.

V. CONCLUSION

In this research paper, the biomechanical analysis of knee and ankle joints of pathological gait was presented. The gait is collected for 22 subjects using Kinect V2 sensor. The observation of the knee and ankle joint curves of one subject was presented. The knee joint has the least actuation when the ankle joint strikes on the ground during the stance phase. On the other side, in the swing phase, the knee joint was highly activated, whereas, ankle joint has the least actuation. The proposed work has validated our hypothesis i.e., knee and ankle joints are inversely related. The calculated joints trajectories are stable for knee and ankle joints.

A. Future Direction

The outcome of this work can be utilized for develop of the wearable device for automatic activity and health monitoring device. It will also help in developing brain control interface device for rehabilitation. The plan of developing the product for automatic gait recognition and reconstruction for analysis of gait abnormality and surveillance.

Ethics Committee Review:

With reference to file no.: ITG/2020/DRD/05/01 dated 15/05/2020, the ethical committee was formulated for verification of all ethical practices. Based on the ethics committee report it is found that all the ethical practice is followed. No Human and animal are harmed during the gait data capturing the process of the stated project work. Name of Committee members: Dr. K.K.S.Mer (Institute Director), Mr. Sachin Sati(Member Secretary), Mr. Vishwnath Bijalwan (Dean R&D), Dr. V.B.Semwal, Mr. Arun Uniyal(member)

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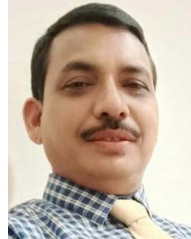


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